

Zachary Nason

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EDUCATION

B.S., Mathematics 2020
University of Kansas, Lawrence, KS

M.S., Mathematics 2023
University of Nebraska-Lincoln, Lincoln, NE

Ph.D., Mathematics Expected May 2027
University of Nebraska-Lincoln, Lincoln, NE
Advisor: Thomas Marley

POSITIONS HELD

Graduate Teaching Assistant August 2021 - Present
Mathematics Department, University of Nebraska-Lincoln

Research Training Group in the Mathematical Sciences (RTG) Graduate Trainee Fall 2024
Mathematics Department, University of Nebraska-Lincoln

AWARDS

Emeritus Faculty Fellowship - Lloyd Jackson Award 2025
This award is given by the UNL Mathematics Department to graduate students to recognize notable scholarship and to promote future mathematical research. This award is named after UNL emeritus professor Lloyd Jackson.

Outstanding Qualifying Exam 2022
This award is given by the UNL Mathematics Department to two graduate students who have achieved the highest scores on the Ph.D. qualifying exams for that year.

PUBLICATIONS/PREPRINTS

- [1] Zachary Nason, Andrew J. Soto Levins, and Ryan Watson. Quasi-Gorenstein morphisms of commutative local dg-algebras. Accessible at <https://arxiv.org/abs/2605.02029>, May 2026.
- [2] Lars Winther Christensen, Antonia Kekkou, Justin Lyle, Zachary Nason, and Andrew J. Soto Levins. G-levels of perfect complexes and a Bass formula for levels. Accessible at <https://arxiv.org/abs/2507.10486>, April 2026.
- [3] Paulo Martins, Victor D. Mendoza Rubio, and Zachary Nason. Finiteness of complete intersection dimensions of RHom complexes and Ext modules. Accessible at <https://arxiv.org/abs/2601.07811>, January 2026.
- [4] Zachary Nason. Maximal Cohen-Macaulay DG-complexes. *Journal of Pure and Applied Algebra*, 230(2):108193, 2026.

TALKS AND PRESENTATIONS

NON-UNL PRESENTATIONS

4. **Maximal Cohen-Macaulay DG-complexes** (50 min) April 13, 2026
University of Texas-Rio Grande Valley, Algebra Seminar
3. **Level inequalities for complexes** (50 min) March 4, 2026
Texas Tech University, Algebra and Number Theory Seminar
2. **Level inequalities for complexes** (50 min) January 30, 2026
Syracuse University, Algebra Seminar
1. **Maximal Cohen-Macaulay DG-complexes and a derived Improved New Intersection Theorem** (20 min) March 29, 2025
*University of Kansas, AMS Spring Central Sectional 2025
Special Session on Topics in Commutative Algebra*

UNL PRESENTATIONS

11. **Koszul property for points in projective space** (50 min) April 6, 2026
University of Nebraska-Lincoln, RTG Learning Seminar: Koszul Algebras
10. **Level inequalities for complexes** (50 min) December 3, 2025
University of Nebraska-Lincoln, Commutative Algebra Seminar
9. **Journeying towards Jensen's theorem** (3 talks, 50 min each) July 21, 23, and 25, 2025
University of Nebraska-Lincoln, Summer Commutative Algebra Reading Seminar
8. **Maximal Cohen-Macaulay DG-modules** (50 min) November 20, 2024
University of Nebraska-Lincoln, Commutative Algebra Seminar
7. **Maximal Cohen-Macaulay complexes and an extension to differential graded rings** (50 min) September 25, 2024
University of Nebraska-Lincoln, Commutative Algebra Reading Seminar
6. **Constructing the derived category** (3 talks, 50 min each) July 29, 31, and August 2, 2024
University of Nebraska-Lincoln, Summer Commutative Algebra Reading Seminar
5. **Cohen-Macaulay differential graded algebras** (50 min) May 1, 2024
University of Nebraska-Lincoln, Commutative Algebra Reading Seminar
4. **Gorenstein differential graded algebras** (50 min) October 19, 2023
University of Nebraska-Lincoln, Commutative Algebra Reading Seminar
3. **Characterizing Gorenstein local rings using the Frobenius map** (50 min) April 19, 2023
University of Nebraska-Lincoln, Commutative Algebra Reading Seminar
2. **Symbolic powers and monomial ideals** (50 min) November 16, 2022
University of Nebraska-Lincoln, Commutative Algebra Reading Seminar
1. **A counterexample concerning the existence of height one prime ideals in the intersection of height two prime ideals** (50 min) February 9, 2022
University of Nebraska-Lincoln, Commutative Algebra Reading Seminar

CONFERENCES ATTENDED

- A Celebration of Golod's Mathematical Legacy April 2026
University of Warwick
- CA+ 2026 March 2026
Iowa State University
- SLAM 2026 March 2026
The University of Texas at Arlington
- KUMUNU October 2025
University of Missouri
- AMS Fall 2025 Southeastern Sectional Meeting October 2025
Tulane University
- Summer CAMP August 2025
University of Nebraska-Lincoln
- Modules and Rings: Recent Developments in Commutative Algebra June 2025
University of Genova
- URiCA May 2025
University of Nebraska-Lincoln
- RayFest April 2025
University of Nebraska-Lincoln
- AMS Spring 2025 Central Sectional Meeting March 2025
University of Kansas
- Commutative Algebra in Mixed Characteristic March 2025
University of Nebraska-Lincoln
- CA+ 2025 March 2025
Iowa State University
- KUMUNU September 2024
University of Missouri
- URiCA May 2024
University of Nebraska-Lincoln
- KUMUNU October 2023
University of Missouri
- AMS Fall 2023 Central Sectional Meeting September 2023
Creighton University
- BrianFest August 2023
University of Nebraska-Lincoln
- URiCA May 2023
University of Nebraska-Lincoln
- Morgantown Algebra Days April 2023
West Virginia University
- Math 125 April 2023
University of Nebraska-Lincoln
- KUMUNU October 2022
University of Nebraska-Lincoln

TEACHING

Recitation Instructor MATH 221: <i>Differential Equations</i>	Spring 2026 (2 sections)
Instructor of Record MATH 103: <i>College Algebra and Trigonometry</i>	Fall 2025
Instructor of Record MATH 107WHT: <i>Calculus II - William H. Thompson Scholars Section</i>	Spring 2025
Instructor of Record MATH 208: <i>Calculus III</i>	Summer 2024
Instructor of Record MATH 104: <i>Applied Calculus</i>	Spring 2024
Instructor of Record MATH 103: <i>College Algebra and Trigonometry</i>	Fall 2023
Grading Assignment MATH 825: <i>Mathematical Analysis I</i>	Fall 2023
Instructor of Record MATH 107: <i>Calculus II</i>	Summer 2023
Algebra Qualifying Exam Review Session Organizer Organized and led the review session for UNL's algebra qualifying exam.	Summer 2023
Recitation Instructor MATH 208: <i>Calculus III</i>	Spring 2023 (3 sections)
Instructor of Record MATH 101: <i>College Algebra</i>	Fall 2022
Recitation Instructor MATH 107: <i>Calculus II</i>	Spring 2022 (2 sections), Summer 2022 (1 section)
Recitation Instructor MATH 106: <i>Calculus I</i>	Fall 2021
Tutoring Drop-in math tutoring center for UNL first-year undergraduates. Tutored for two hours per week, one semester per year.	Fall 2021-Present

SERVICE

Commutative Algebra Reading Seminar Co-Organizer <i>University of Nebraska-Lincoln</i> Organized and led a weekly graduate student seminar in commutative algebra along with two other graduate students.	2025-Present
UNL DRP Mentor <i>University of Nebraska-Lincoln</i> Mentored an advanced undergraduate student in basic commutative algebra and planar algebraic geometry, with the eventual goal of proving Bezout's Theorem for projective curves in projective 2-space	2024-2025
UNL Graduate Mentor <i>University of Nebraska-Lincoln</i> Mentored a beginning mathematics graduate student during his first year of graduate school. Advised and helped the first-year graduate students with navigating academic life and with adjusting to Lincoln.	2023-2024

UNL Math Day Volunteer

2021-Present

University of Nebraska-Lincoln

The mathematics department at the University of Nebraska-Lincoln holds an annual math day for high school students in Nebraska. Students compete in a math bowl and participate in mathematical activities and puzzle-solving.

COMPUTER SKILLS

Languages: Java, Python, C++

Software: $\text{\LaTeX}$, Macaulay2, MATLAB, Microsoft Office, Google Workspace